Personalized Tourism Management System

A Simple, No-Jargon Guide: How It Works, Who Built What, and What Every File Does

What is this project in 1 sentence?
It is a desktop app (like a mini MakeMyTrip or Expedia) built with Java and MySQL where people can search holiday trips, get smart recommendations based on their budget, book tickets, download PDF invoices, and see airline boarding passes, while an admin manages the trips and looks at sales graphs.

1. The Big Picture: How Does the Code Work?

Think of the application as a restaurant with three main teams working together:

1. The Waiter (The UI - User Interface):
These are the screens and buttons the user sees (the login window, the cards showing pictures of Bali and Paris, the booking pop-up). The UI doesn't do math or touch the database directly; it just takes orders from the user and hands them to the kitchen.

2. The Chef (The Services - Business Logic):
This is the brain of the app. It checks: Is the user's password correct? Does the trip match their budget? How much is 80% if they cancel? Is their credit card valid? It does all the thinking and calculations.

3. The Store Room (The DAOs & Database):
DAO stands for 'Data Access Object'. These files act like storekeepers. Whenever the Chef needs recipes or ingredients (like trip details or customer accounts), the DAO runs a secure SQL command to pull it from or save it to MySQL.

2. The Logic Behind the Application (In Simple Terms)

Here are the 4 main logical rules that power the application:

Rule 1: How the Smart Recommendation Engine Works (Gowri P)

When a user says: *'I have Rs. 50,000, I want a 5-day Beach holiday for 2 people'*, the app looks at all trips in the database and gives each trip a match score between **0% and 100%** using three questions:

- **Does it fit your budget? (40% weight):** It calculates the total cost (ticket price + daily food/stay allowance). If it fits your budget, you get full points. If it's too expensive, points are deducted. If it is more than 30% over your budget, this score drops to 0.
- **Does it fit your time? (35% weight):** If you asked for 5 days and the package is 5 days, you get 100%. If the package is 4 days or 6 days, you get partial points. If it's way too long or short, points are reduced.
- **Does it match what you like? (25% weight):** If you like 'Beach' and the trip is Goa or Bali, you get 100% for this test. If it's a Mountain trip, you get only 15%.

The app adds up the weighted scores, sorts the trips from highest to lowest, and shows the best ones first!

Rule 2: How Prices & Refunds Work (Garry George)

- **Calculating Total Price:** Total = (Ticket Price x Number of Guests) - Any Discount. For example, if a trip to Bali is Rs. 10,000 per person with a 10% discount, 2 people pay Rs. 18,000 instead of Rs. 20,000.
- **Cancellation & Refund Rule:** If a customer cancels their booking **more than 48 hours (2 days)** before the trip starts, they automatically get an **80% refund**. If they cancel less than 48 hours before, or after the trip has already

started, they get **0% refund**.

Rule 3: Password Security (Amy)

When you type a password like secret123, the app **never** saves secret123 in the database. Instead, it scrambles it using a tool called **BCrypt** into a 60-character scrambled code (e.g. $2a$12$ARX4KA...). Even if an attacker stole the entire database, they would never be able to read anyone's real password.

Rule 4: Multi-Currency & Digital Boarding Pass (Amy)

The app has a currency dropdown (USD $, EUR €, GBP £, INR ■, JPY ¥). When you change the currency, every card on the screen instantly recalculates its price into that currency without having to refresh or restart. Also, when you confirm a trip, you can open a digital boarding pass that looks like a real flight ticket with a seat number, gate, and drawn barcode.

3. Who Built What? (Team Work Breakdown)

Here is what every member of the project team did, in plain everyday terms:

Team Member	What They Were Responsible For	What They Actually Built
Amy (Project Lead & Architect)	Set up the foundation, connected everyone's code, built user login, security, and main dashboard.	• Created the project skeleton and database schema (6 tables). • Built the Login and Register screens with password encryption. • Built the main windows for Customer and Admin. • Styled the whole app in warm sunset terracotta colors. • Added real-time currency conversion (USD, INR, EUR, etc.). • Created the digital airline boarding pass with barcodes. • Reviewed everyone's code and merged all pull requests.
Aein Shaji (Dev 3)	Trip Catalog & Package Cards	• Created the screen where users browse all travel packages. • Built the filter buttons (Beach, Adventure, Culture, Nature, Luxury). • Designed the travel cards showing photos, trip duration, price, and the 'Book Now' button.
Gowri P (Dev 4)	Smart Recommendations & Day-by-Day Itinerary	• Created the Recommendation algorithm that calculates a 0-100% match score. • Built the survey screen where users input their budget, days, and travel style. • Created the popup window showing Day 1, Day 2, Day 3 activity schedules for each trip.
Garry George (Dev 5)	Booking, Payments & Cancellation	• Built the 'Book Trip' popup window with date picker and guest counter. • Created the Credit Card and UPI payment options. • Programmed the 48-hour 80% refund logic when someone cancels. • Built the 'My Bookings' screen where users see their booked trips.
Pius Varghese (Dev 6)	Admin Inventory & Booking Oversight	• Built the Admin screen where managers can Add, Edit, or Delete trips. • Built the master table where admins can see every booking made by every customer.
Ashwin V (Dev 7)	PDF Downloads & Admin Sales Charts	• Integrated the PDF generator to download Itineraries and Tax Invoices. • Built the Admin Reports screen with colorful bar charts and pie charts showing top destinations and monthly revenue.

Atul (Dev 8)	Reviews & Trip Alerts	• Built the 5-star rating and review popup for customers. • Built the alert banner on the customer dashboard reminding them if their trip is coming up in the next 7 days.

4. What Does Each File Do? (Simple Cheat Sheet)

Here is what each file in the project folder does, grouped by folder:

Folder & File Name	What It Does In Plain Words
config / DBConnection.java	Connects Java to MySQL. Ensures the app uses one shared connection instead of opening hundreds of connections.
config / ThemeConstants.java	The stylebook. Stores the colors (warm terracotta, coral, white), fonts, and button sizes so the whole app looks consistent.
config / CurrencyManager.java	Handles money conversion (USD, INR, EUR, GBP, JPY) so prices change live on screen.
config / LanguageManager.java	Handles translations (English, Spanish, French, German, Hindi, Japanese).
model / User.java	A box holding user information: user ID, full name, email, role (Customer or Admin), and phone.
model / Destination.java	A box holding destination details: city name, country, description, and daily spending cost.
model / Package.java	A box holding trip package details: title, number of days, price, and discount.
model / ItineraryDay.java	A box holding one day's schedule (e.g. Day 1: Arrive and hotel check-in).
model / Booking.java	A box holding a booked ticket: travel date, guests, total amount paid, and status (CONFIRMED or CANCELLED).
model / Review.java	A box holding customer feedback: 1 to 5 stars and written review.
model / CustomerBookingDossier.java	A combined report used by admins that shows a customer's name, the trip they picked, and their masked payment info.
model / UserSession.java	Remembers who is currently logged in so you don't have to re-enter your password on every screen.
dao / UserDAO.java	Saves new users to MySQL and finds users by email when they log in.
dao / DestinationDAO.java	Fetches destination places (like Bali, Swiss Alps, Kyoto) from MySQL.
dao / PackageDAO.java	Fetches all travel packages from MySQL and handles search filters.
dao / ItineraryDAO.java	Fetches the Day 1, Day 2, Day 3 activity schedule for any trip package.
dao / BookingDAO.java	Saves bookings into the database and marks them as cancelled if requested.
dao / ReviewDAO.java	Saves customer reviews and calculates average ratings for each package.
dao / AdminDAO.java	Runs special administrative queries to list all customer dossiers and payment states.
service / AuthService.java	Checks if the entered email and password match, and scrambles new passwords with BCrypt.
service / RecommendationService.java	The matching brain. Scores packages from 0% to 100% based on user budget, duration, and preferences.
service / ItineraryService.java	Retrieves and arranges the day-wise itinerary activities in order.
service / BookingService.java	Checks booking dates, calculates final price, checks payment, and calculates the 80% refund if cancelled early.
service / CardPayment.java	Checks credit card numbers (must be 16 digits), checks expiration date, and hides all but the last 4 digits for safety.
service / UpiPayment.java	Checks UPI IDs (must be in the format name@bank) and generates a payment confirmation ID.

service / PDFExportService.java	Builds downloadable PDF files for the travel itinerary and the official tax invoice.
service / AnalyticsService.java	Calculates total revenue and most popular destinations so the admin can see graphs.
Main.java	The starting point of the app. Starts the FlatLaf modern theme, checks database connection, and opens the Login screen.
ui / auth / LoginFrame.java	The login screen with email, password, and links to register.
ui / auth / RegisterFrame.java	The registration screen for new customers to create an account.
ui / customer / CustomerMainFrame.java	The main dashboard for travelers with tabs for Catalog, Recommendations, and My Bookings.
ui / customer / CatalogPanel.java	The trip browsing screen with search bar and category filter chips.
ui / customer / PackageCard.java	The visual card showing trip image, live weather badge, star rating, price, and 'Book Now' button.
ui / customer / RecommendationsPanel.java	The preferences survey where users select their budget and get matched trips.
ui / customer / ItineraryDialog.java	The popup window showing the day-by-day plan with a 'Download PDF' button.
ui / customer / BookingDialog.java	The popup window where customers choose their travel date, guests, and pay via Card or UPI.
ui / customer / BookingHistoryPanel.java	The screen listing your booked trips with buttons to cancel, download invoice, or get a boarding pass.
ui / customer / BoardingPassDialog.java	A stylish digital flight ticket with gate number, seat assignment, and barcode.
ui / customer / ReviewDialog.java	The popup window where customers give 1 to 5 stars and leave comments.
ui / admin / AdminMainFrame.java	The main dashboard for the system administrator.
ui / admin / InventoryManagementPanel.java	The admin screen to Add, Edit, or Delete trips in the database.
ui / admin / BookingOversightPanel.java	The master table where admins monitor all bookings made by customers.
ui / admin / CustomerBookingsAdminPanel.java	Admin table showing customers, their chosen trips, and masked payment settlements.
ui / admin / CustomerDossierDialog.java	Admin popup showing a detailed profile and booking history of a specific customer.
ui / admin / ReportsPanel.java	The analytics screen showing colorful JFreeChart bar graphs and revenue charts.
resources / tourism_schema.sql	The SQL script that creates all 6 tables and fills them with default trips for Bali, Paris, Swiss Alps, Kyoto, Manali, Goa, Shimla, Kerala, Vietnam, Maldives, and Egypt.

5. How to Run the App (Quick Steps)

Step 1: Start MySQL on your computer.
Step 2: Open terminal in the project folder and run:

```
mvn clean compile
mvn exec:java
```

Step 3: Log in with either of the test accounts:

- **Customer Account:** Email: john@gmail.com | Password: user123
- **Admin Account:** Email: admin@tourism.com | Password: admin123